

Intelligence Leading the Frontier, Computing Shaping the Future HIT Code Review Research

Experiment Report Mining Code Review Comments and Understanding Requirements

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Lab Score		Discussion Score	
Report Score		Total Experi- ment Score	

1. Experiment Objectives

- Understood traditional machine learning workflow for code review. - Mastered data preprocessing for code review tasks. - Extracted AST and CFG representations from code. - Extracted structural, textual, and statistical features. - Trained SVM and Random Forest models. - Built a merge prediction model and analyzed performance.

2. Experiment Content

- Filtered 1,020 human-written PRs from Experiment 1 dataset. - Extracted AST representations using Python's ast module. - Extracted CFG representations for control flow analysis. - Extracted 9 features: additions, deletions, changed_files, commits_count, reviewers_count, comments_count, labels_count, pr_length, has_ai_reviewer. - Constructed ML dataset: 816 train, 204 test. - Trained SVM (RBF kernel) and Random Forest (100 trees). - Evaluated using accuracy, precision, recall, F1, ROC-AUC.

3. Experimental Results

We trained SVM and Random Forest on 1,020 human-written PRs (601 merged, 419 not merged).

SVM Results: - Accuracy: 88.7% - Precision: 88.2% - Recall: 93.3% - F1: 90.7% - ROC-AUC: 92.3%

Random Forest Results: - Accuracy: 91.2% - Precision: 88.6% - Recall: 97.5% - F1: 92.9% - ROC-AUC: 97.3%

Random Forest outperformed SVM across all metrics. Feature importance analysis showed labels_count (45.3%) as the dominant predictor, followed by pr_length (12.2%) and deletions (10.3%).

AST and CFG examples were generated for sample code snippets.

Figures

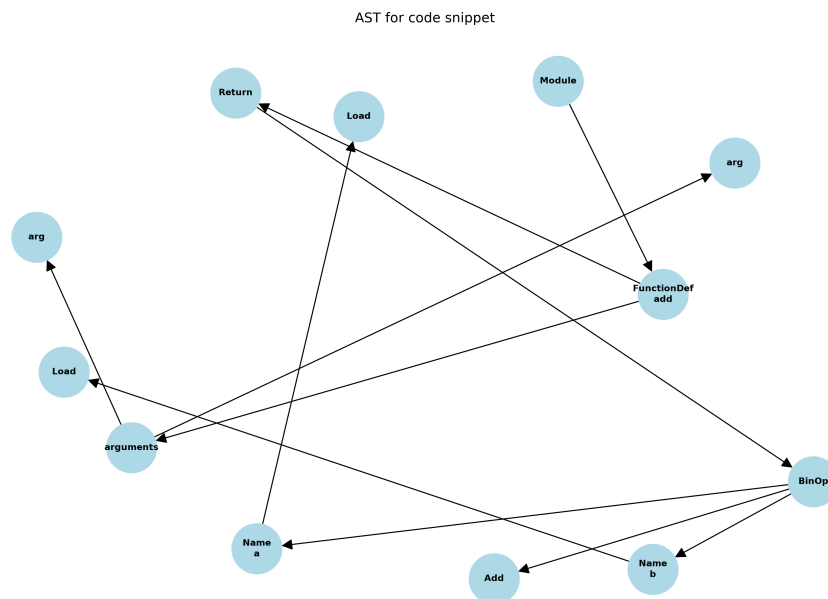


Figure 1: AST for a simple addition function

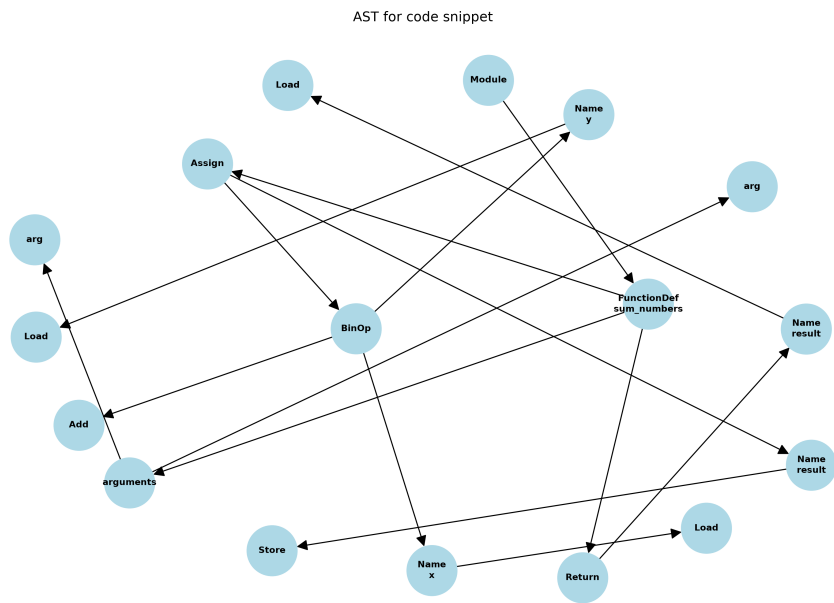


Figure 2: AST for a function with intermediate variable

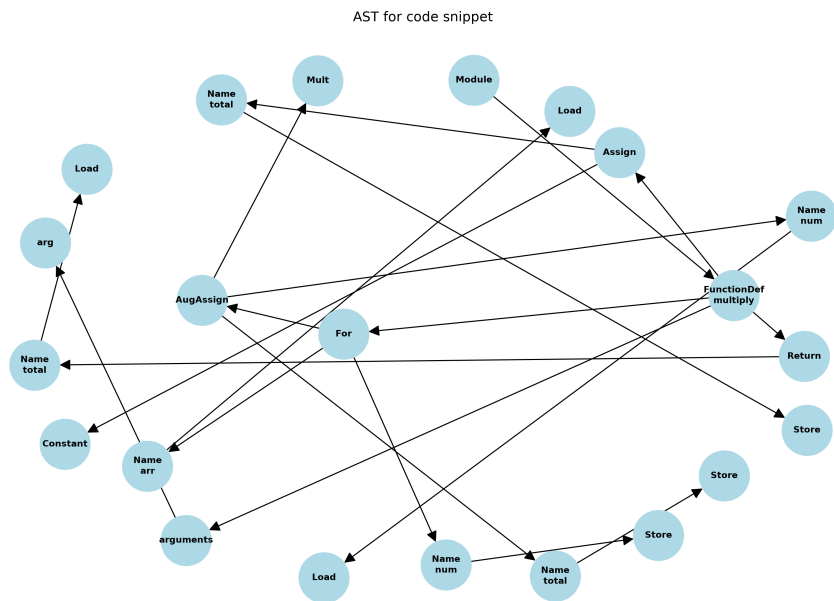


Figure 3: AST for a function with loop

CFG for code snippet

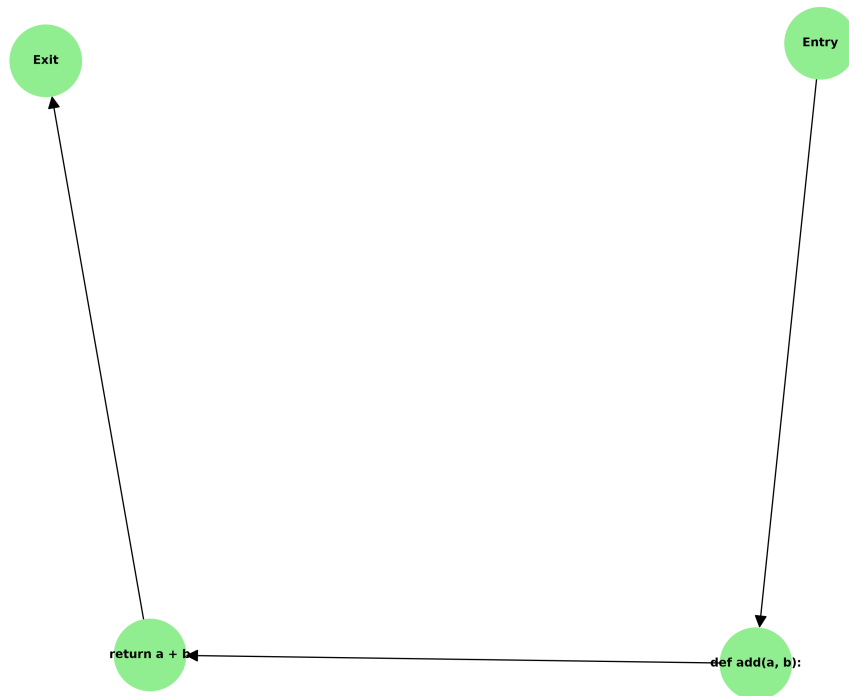


Figure 4: CFG for a simple addition function

CFG for code snippet

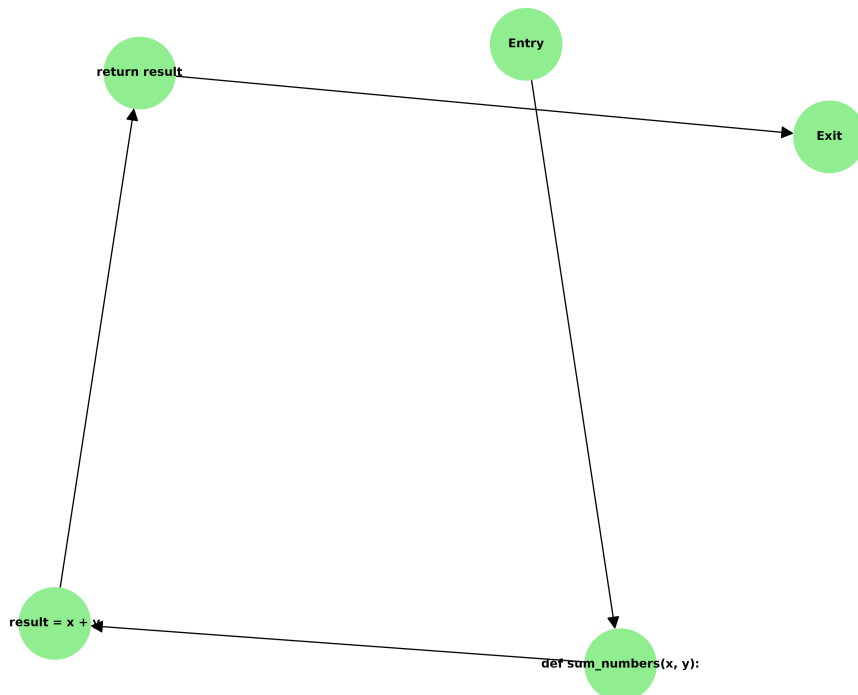


Figure 5: CFG for a function with intermediate variable

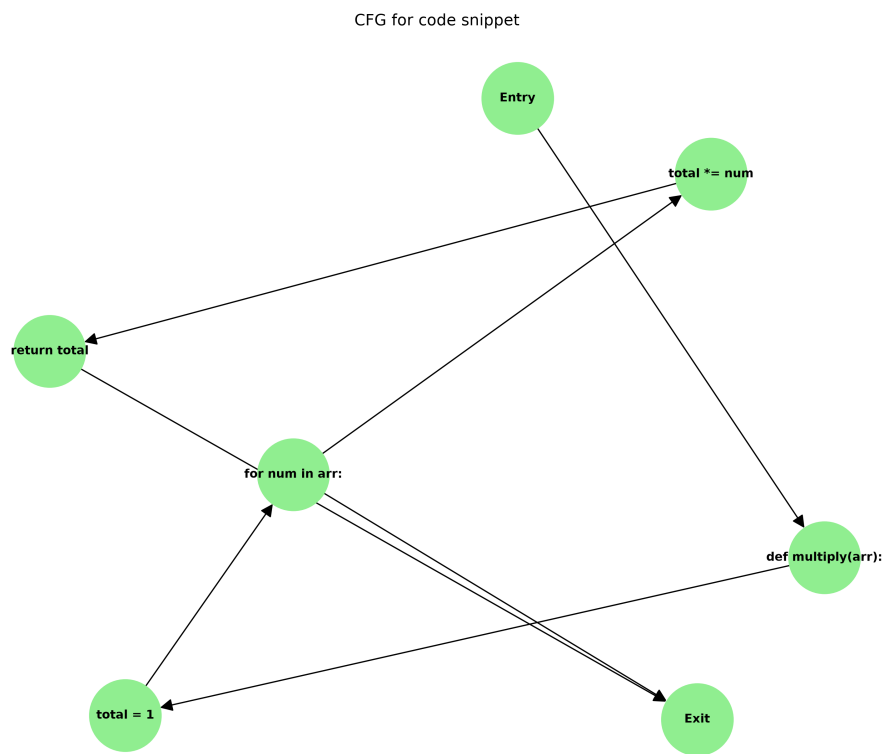


Figure 6: CFG for a function with loop

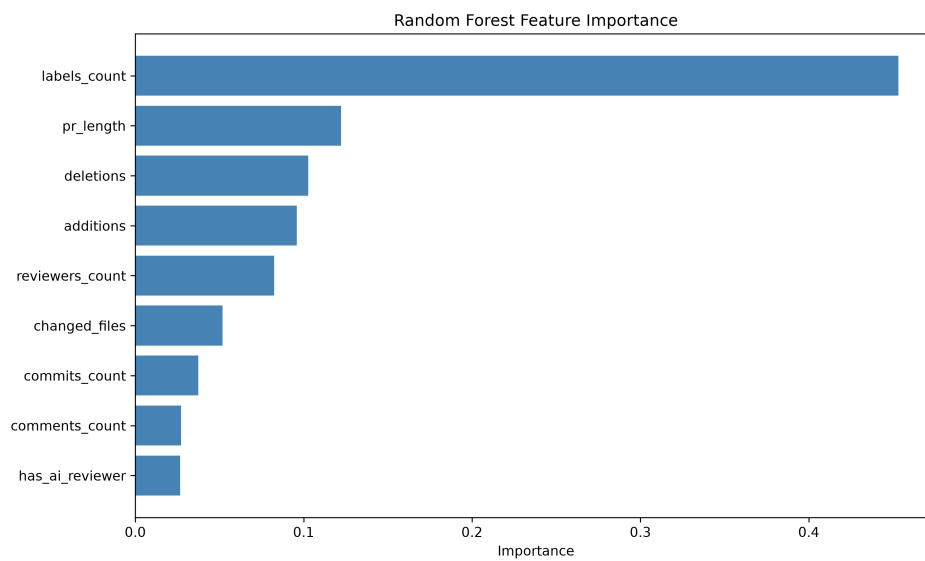


Figure 7: Random Forest feature importance

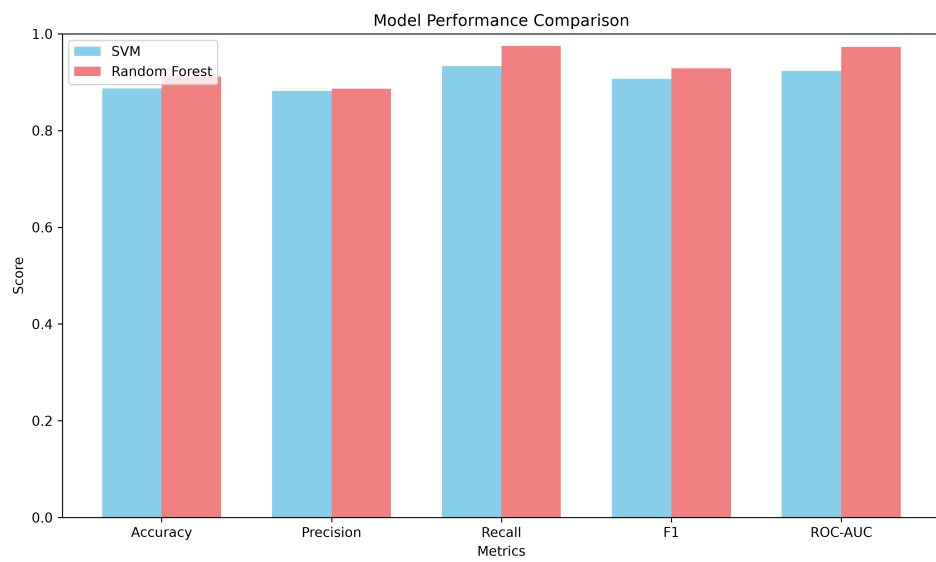


Figure 8: SVM vs Random Forest performance comparison

4. Summary of Problems Encountered

- Pandas and scikit-learn missing in venv - installed via pip. - AST/CFG visualization required networkx - installed separately. - sklearn package name deprecated - used scikit-learn instead.

5. Reflections on the Experiment

- AST/CFG capture program semantics that raw text cannot represent, enabling deeper code understanding. - Traditional ML requires manual feature engineering because algorithms cannot directly interpret code syntax and semantics. - SVM works well for small datasets with clear separation; Random Forest handles complex interactions and provides feature importance. - labels_count was the most important feature - PRs with more labels are better organized and more likely to merge. - Experiment 2 added feature engineering, scaling, model training, and evaluation to the data collection pipeline.

Instructor's Comments: Date: